CALCULUS

Chapter 15

Derivative Applications

15

SYLLABUS 5.8 5.9 5.14

Throw a ball straight up. It rises, it slows, and at the very top it stops for an instant before it falls back. At that instant its velocity is exactly zero. A moment earlier the ball was moving up; a moment later it is moving down. The top of the flight is the one place where it is doing neither.

That instant is the key to a large class of problems. The top of the flight is where the height is greatest, and it is also where the velocity is zero. The same pattern holds everywhere: when a quantity reaches its largest or smallest value, its rate of change is zero at that moment, and the curve flattens.

This is why the derivative is the right tool for finding a best value. The largest volume from a fixed sheet of metal, the least metal for a can of a given size, the quickest route, the cheapest design: in each case the answer sits where the derivative is zero. In this chapter the derivative stops being something you calculate and becomes something you use. You will find the peaks and valleys of a curve and decide which is which, solve optimisation problems, describe motion, and work with quantities that change together.

The ball at the top of its flight holds the whole idea: to find an extreme value, find where the rate of change is zero.

By the end of this chapter you will be able to find and classify the stationary points and points of inflexion of a curve, and to read its shape from the signs of f' and f''. You will use a derivative to find a largest or smallest value under a constraint, describe motion in a straight line by differentiating displacement, and differentiate implicitly when two rates are linked. The course spends this long on it because a derivative is not only a gradient. It is a rate of change, and many real questions ask how fast something is changing, or where a quantity is greatest or least.

15.1 Stationary Points and Their Nature

A **stationary point** is a point where the gradient is zero: the tangent is horizontal. Since the derivative gives the gradient, stationary points are exactly the solutions of $f'(x) = 0$. They are the points where a curve stops rising or stops falling.

For a smooth curve, every local maximum and minimum is a stationary point. The condition $f'(x) = 0$ finds them all. It does not find an extreme value at a corner, where the derivative does not exist: $f(x) = |x|$ has a minimum at $x = 0$ and no stationary point there. See Ch. 14.

DEF A **stationary point** of f occurs where $f'(x) = 0$. At a **local maximum** the curve changes from increasing to decreasing; at a **local minimum**, from decreasing to increasing.

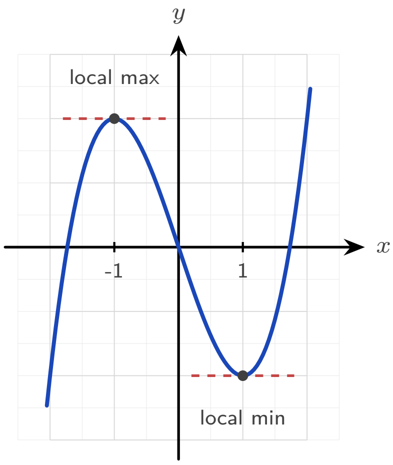

Figure 15.1 At each stationary point the tangent is horizontal. The second derivative decides which is the maximum and which the minimum.

Finding stationary points is only half the work. The other half is deciding what kind each one is: a maximum, a minimum, or neither. There are two tests.

The first derivative test

The sign of f' tells you whether the curve is rising or falling, so it can be read on either side of the stationary point. Just before a maximum the curve rises, $f' > 0$, and just after it falls, $f' < 0$. A minimum is the reverse. Record the three signs in order and the shape is decided.

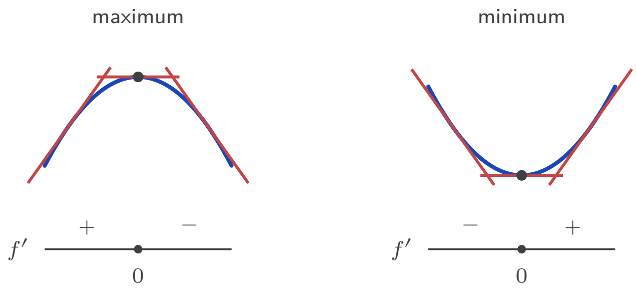

Figure 15.2 The first derivative test. The scarlet strokes are tangents: rising, flat, then falling at a maximum, and the reverse at a minimum. The strip below records the sign of f' on each side.

The second derivative test

This test is usually faster, and it uses the second derivative, which measures concavity. There is a picture behind it. At a minimum the curve bends upward, so it sits *above* its own horizontal tangent, and $f'' > 0$. At a maximum it bends downward, sits *below* the tangent, and $f'' < 0$.

Figure 15.3 The second derivative test. A curve that bends upward lies above its horizontal tangent, and one that bends downward lies below it. That is what the sign of f'' records.

KEY At a stationary point where $f'(x) = 0$:

$$f''(x) > 0 \Rightarrow \text{local minimum}, \qquad f''(x) < 0 \Rightarrow \text{local maximum}.$$

If $f''(x) = 0$ this test gives no answer. Use the sign of f' on either side instead.

EXAMPLE 15.1

Find and classify the stationary points of $f(x) = x^3 - 3x^2 - 9x + 5$.
Differentiate and solve $f'(x) = 0$:

$$f'(x) = 3x^2 - 6x - 9 = 3(x-3)(x+1) = 0 \implies x = -1 \text{ or } x = 3.$$

The second derivative is $f''(x) = 6x - 6$.
At $x = -1$: $f''(-1) = -12 < 0$, a local maximum. $f(-1) = 10$, so $(-1, 10)$.
At $x = 3$: $f''(3) = 12 > 0$, a local minimum. $f(3) = -22$, so $(3, -22)$.
Here the maximum is above the minimum, which is normal. The word "local" means highest or lowest *among nearby points*, not over the whole curve.

YOUR TURN 15.1 Stationary Points and Their Nature

1. Find the stationary points of each curve.

(a) $y = x^2 - 6x + 5$ [2]
(b) $y = x^3 - 12x$ [2]
(c) $y = 2x^3 - 3x^2 - 12x + 1$ [3]
(d) $y = x^4 - 2x^2$ [3]

2.

(a) Use the second derivative to classify the stationary point of $y = x^2 - 6x + 5$. [2]
(b) Use the second derivative to classify both stationary points of $y = x^3 - 12x$. [3]
(c) Find the stationary point of $y = x + \dfrac{1}{x}$ for $x > 0$, and state its nature. [3]
(d) A curve has $\dfrac{dy}{dx} = (x-1)(x-4)$. Write down the x-coordinates of its stationary points, and state the nature of each without differentiating again. [4]

3. The curve $y = x^3$ has a stationary point at the origin.

(a) Show that $x = 0$ is a stationary point. [2]
(b) Explain why the second derivative test cannot classify it. [2]
(c) Use the sign of $\dfrac{dy}{dx}$ on either side to classify it. [3]

15.2 Points of Inflexion and Concavity

A stationary point is not the only feature worth naming. A **point of inflexion** is where the curve changes the *direction of its bend*: from concave up to concave down, or the reverse. It is where f'' changes sign, which requires $f''(x) = 0$.

DEF A **point of inflexion** is a point where the concavity changes sign. A necessary condition is $f''(x) = 0$, but $f''(x) = 0$ alone is not sufficient: the concavity must actually change.

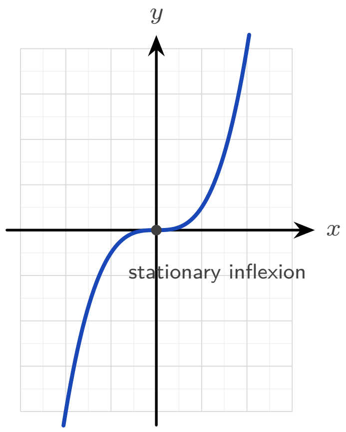

Figure 15.4 At the origin $y = x^3$ has both a horizontal tangent and a change of concavity, so this inflexion is also a stationary point.

An inflexion can happen at any gradient, and the syllabus names both kinds.

In $y = x^3$ above, the inflexion at the origin has **zero gradient**. The tangent is horizontal, so this point is also a stationary point, but it is neither a maximum nor a minimum.

In the next example the inflexion has a **non-zero gradient**. The curve is falling as it passes through.

Only one thing makes a point an inflexion: the concavity changes there. The gradient can take any value.

EXAMPLE 15.2

Find the point of inflexion of $f(x) = x^3 - 3x^2$.
Differentiate twice:

$$f'(x) = 3x^2 - 6x, \qquad f''(x) = 6x - 6.$$

Set $f''(x) = 0$: $6x - 6 = 0 \implies x = 1$. To confirm it is genuine, check that f'' changes sign: for $x < 1$, $f'' < 0$ (concave down); for $x > 1$, $f'' > 0$ (concave up). It changes, so $x = 1$ is a point of inflexion. Since $f(1) = -2$, the point is $(1, -2)$.
Note the gradient there: $f'(1) = 3 - 6 = -3 \neq 0$. This is a *non-stationary* inflexion, the curve changing its bend mid-descent.

CAUTION *$f''(x) = 0$ does not by itself guarantee an inflexion. For $f(x) = x^4$, $f''(0) = 0$, yet the curve is concave up on both sides: $x = 0$ is a minimum, not an inflexion. Always confirm the concavity actually changes.*

YOUR TURN 15.2 Points of Inflexion and Concavity

4. Find the point of inflexion of each curve.

(a) $y = x^3 - 6x^2 + 5$ [2]
(b) $y = x^3 + 3x^2$ [2]
(c) $y = 2x^3 - 9x^2 + 12x$ [2]

5.

(a) Explain why $y = x^4$ has no point of inflexion at $x = 0$. [2]
(b) Determine, for $y = x^5$, whether the point where $f'' = 0$ is a genuine inflexion. [3]
(c) State the interval on which $y = x^3 - 6x^2 + 5$ is concave down. [2]

15.3 Optimization

Optimisation means finding the largest or smallest value of a real quantity: the greatest volume, the least cost, the shortest time.

It is worth asking why the derivative helps at all. Without it, the only way to find a best value would be to try inputs one at a time and compare the results. That can suggest an answer but never settles it, because between any two inputs you tried there are infinitely many you did not. The derivative removes the search. A maximum or a minimum leaves a mark that algebra can detect: the curve is flat there, so $f'(x) = 0$. An impossible hunt through infinitely many values becomes one equation to solve, and the handful of solutions are the only candidates worth checking. That is the whole reason this chapter exists.

In practice the calculus is the easy part; the work is in setting up the function.

Almost every optimisation question follows the same five steps.

1. **Name the quantity to be optimised** and write an expression for it. At this stage it may contain more than one variable.
2. **Find the constraint**, the second equation linking those variables. It is always in the question, often as a fixed volume, a fixed length of material, or a fixed perimeter.
3. **Use the constraint to eliminate variables**, until the quantity is a function of one variable only. State the values that variable may take.
4. **Differentiate, set the derivative to zero, and solve.**
5. **Justify the nature of the point**, then answer the question that was actually asked.

Steps 3 and 5 are the ones to watch. Reducing to a single variable is where the algebra goes wrong, and skipping the justification is where marks are dropped at the end.

TIP *An optimisation question is not finished when you solve $f'(x) = 0$. Two marks usually depend on what follows: justify the nature of the point, using f'' or the sign of f', and then answer the question that was asked. If the question asks for the maximum volume, give the volume, not the value of x that produces it.*

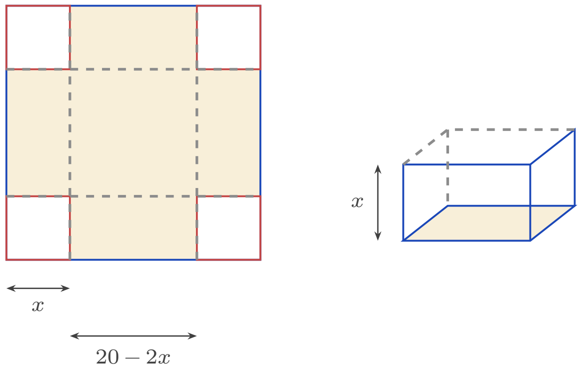

Figure 15.5 Cutting a square of side x from each corner of the card, then folding the flaps up. The base of the box measures $20 - 2x$ each way and its height is x, so $V = x(20 - 2x)^2$.

EXAMPLE 15.3

An open box is made from a $20{\times}20$ cm square of card by cutting a square of side x from each corner and folding up the sides. Find the value of x that maximises the volume.
After cutting, the base is $(20 - 2x)$ by $(20 - 2x)$ and the height is x, so

$$V(x) = x(20 - 2x)^2, \qquad 0 < x < 10.$$

Differentiate (expanding the bracket helps):

$$V'(x) = (20 - 2x)^2 + x \cdot 2(20 - 2x)(-2) = (20 - 2x)(20 - 6x).$$

Setting $V'(x) = 0$ gives $x = 10$ (rejected, no box) or $x = \frac{10}{3}$.
Since $V' > 0$ before $\frac{10}{3}$ and $V' < 0$ after, this is a maximum. The maximum volume is

$$V\left(\tfrac{10}{3}\right) = \frac{16000}{27} \approx 593 \text{ cm}^3.$$

The restriction $0 < x < 10$ comes from the geometry. If x is 0 the box has no height; if x is 10 it has no base.

EXAMPLE 15.4

A closed cylindrical can must hold $355\,\text{cm}^3$. Find the radius that minimises the surface area.
Volume fixes the height: $\pi r^2 h = 355$, so $h = \dfrac{355}{\pi r^2}$. The surface area is two circles plus the curved side:

$$S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \frac{710}{r}.$$

Differentiate and set to zero:

$$\frac{dS}{dr} = 4\pi r - \frac{710}{r^2} = 0 \implies r^3 = \frac{710}{4\pi} \implies r \approx 3.84 \text{ cm}.$$

Since $\frac{d^2S}{dr^2} = 4\pi + \frac{1420}{r^3} > 0$, this is a minimum. Using the constraint to remove h is the step that reduces two variables to one.

CAUTION *Setting $f' = 0$ finds only the best values inside an interval. If the variable is restricted, for example $0 < r \leq 3$, the best value may sit at an* ***endpoint*** *instead, where the derivative need not be zero.* HL *Always list the candidates: every stationary point in the interval, and both endpoints. Then compare their values and take the largest or the smallest.*

EXAMPLE 15.5

For the can above, the manufacturer's machinery cannot roll a radius above 3 cm, so $0 < r \leq 3$. Find the radius that minimises the surface area now.

The unconstrained minimum at $r \approx 3.84$ is no longer allowed. Consider the derivative on $(0, 3]$:

$$\frac{dS}{dr} = 4\pi r - \frac{710}{r^2}.$$

Both terms increase as r increases, so $\frac{dS}{dr}$ is increasing on this interval. Testing the largest value is therefore enough: at $r = 3$, $12\pi - \frac{710}{9} \approx -41 < 0$. The derivative is negative at the right-hand end, and smaller still to the left, so it is negative throughout and S is decreasing across the whole interval. The minimum therefore sits at the endpoint $r = 3$, with $S = 18\pi + \frac{710}{3} \approx 293$ cm^2, even though $S'(3) \neq 0$. Whenever the domain is restricted, check its endpoints as well. The condition $f' = 0$ finds only the best values that lie *inside* the interval.

YOUR TURN 15.3 Optimization

6.

(a) Two numbers have a sum of 20. Find the maximum value of their product, and justify that it is a maximum. [4]

(b) A rectangle has a perimeter of 40 cm. Find the dimensions giving the maximum area. [4]

(c) Find the minimum value of $S = x^2 + \frac{16}{x}$ for $x > 0$, justifying that it is a minimum. [4]

7. An open box is made from a 12×12 cm square of card by cutting a square of side x cm from each corner and folding up the sides.

(a) Show that the volume is $V = x(12 - 2x)^2$, and state the possible values of x. [3]

(b) Find the value of x that maximises the volume. [4]

(c) Find the maximum volume. [2]

8.

(a) A rectangular field is fenced on three sides, the fourth being a wall, using 100 m of fencing. Find the maximum area. [4]

(b) An open-topped cylindrical tank must hold 1000 cm^3. Show that its surface area is $S = \pi r^2 + \dfrac{2000}{r}$, and find the radius that minimises it. [5]

(c) For the tank in part (b), explain why the answer would change if the manufacturer could not make a radius larger than 5 cm. [3]

15.4 Kinematics

Motion is change of position, so calculus describes it exactly. If a particle's **displacement** from a fixed origin is $s(t)$, then its **velocity** is the rate of change of displacement, $v = \dfrac{ds}{dt}$, and its **acceleration** is the rate of change of velocity.

KEY · IB

$$a = \frac{dv}{dt} = \frac{d^2 s}{dt^2}$$

Velocity carries a sign, so it records direction as well as how fast the particle moves. **Speed** is the size of the velocity, written $|v|$, and it is never negative.

The particle is **momentarily at rest** (at rest for an instant) when $v = 0$. This is the stationary-point condition of §15.1, applied to displacement.

Speed follows a separate rule. The particle **speeds up when v and a have the same sign**, because the acceleration acts in the direction of motion. It slows down when the two signs are different.

One quantity needs integration rather than differentiation: the **total distance travelled**, which keeps counting when the particle turns and comes back. See Ch. 16.

EXAMPLE 15.6

A particle moves so that its displacement is $s(t) = t^3 - 9t^2 + 24t$ metres, for $t \geq 0$ seconds. Find when it is at rest, and its acceleration at those times.

Differentiate for velocity:

$$v(t) = 3t^2 - 18t + 24 = 3(t-2)(t-4).$$

The particle is at rest when $v = 0$: at $t = 2$ and $t = 4$ seconds.

Acceleration is $a(t) = 6t - 18$. At $t = 2$, $a = -6\,\text{m}\,\text{s}^{-2}$; at $t = 4$, $a = 6\,\text{m}\,\text{s}^{-2}$. The sign change of velocity at each rest point means the particle reverses direction there.

> Now ask whether the particle is speeding up at $t = 5$. There $v(5) = 3(3)(1) = 9 > 0$ and $a(5) = 6(5) - 18 = 12 > 0$. The two signs are the same, so the particle is speeding up: it moves in the positive direction and the acceleration acts the same way.

TIP *In kinematics questions, state the units with every answer:* $\mathrm{m\,s^{-1}}$ *for velocity,* $\mathrm{m\,s^{-2}}$ *for acceleration. A correct number without its unit loses the accuracy mark.*

CAUTION *Velocity and speed are not the same. A particle with* $v = -5$ *moves in the negative direction at a speed of* 5*. Maximum speed can therefore occur where the velocity is most negative, so compare* $|v|$*, not* v*.*

YOUR TURN 15.4 Kinematics

9. A particle moves in a straight line.

(a) Its displacement is $s = t^2 - 4t + 3$. Find when it is at rest, and its acceleration then. [2]
(b) Its displacement is $s = 2t^3 - 9t^2 + 12t$. Find the times when it is at rest. [3]
(c) Its displacement is $s = t^3 - 3t^2$. Find its velocity and acceleration at $t = 2$. [2]
(d) Its velocity is $v = t^2 - 5t + 6$. Find when it is at rest, and its acceleration at those times. [3]

10. A particle has velocity $v = 3t^2 - 12t + 9$ for $t \geq 0$.

(a) Find the times when it is at rest. [2]
(b) Find its acceleration at those times. [2]
(c) Determine whether the particle is speeding up or slowing down at $t = 1.5$, giving a reason. [3]
(d) A second particle has $v = t - 4$ and $a = 1$ at $t = 2$. State, with a reason, whether it is speeding up or slowing down. [2]

15.5 Implicit Differentiation HL

So far every curve has been given as a function $y = f(x)$, with y written on its own. But an equation such as $x^2 + y^2 = 25$ also defines a curve, without y being isolated.

Implicit differentiation finds the gradient of such a curve directly, without first solving for y.

The method is the chain rule. Treat y as an unknown function of x.

A term in y then differentiates as usual, but it gains an extra factor $\dfrac{dy}{dx}$, because you are differentiating a function of y with respect to x. For example, $\dfrac{d}{dx}(y^2) = 2y\dfrac{dy}{dx}$.

EXAMPLE 15.7

Find $\dfrac{dy}{dx}$ for the circle $x^2 + y^2 = 25$, and the gradient at $(3, 4)$.
Differentiate both sides with respect to x:

$$2x + 2y\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}.$$

At $(3, 4)$ the gradient is $-\frac{3}{4}$. The answer depends on both coordinates, and it must: a circle has a different gradient at every point.

EXAMPLE 15.8

Find $\dfrac{dy}{dx}$ for $x^2 + xy + y^2 = 7$.
The middle term needs the product rule. Differentiating term by term:

$$2x + \left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0.$$

Collect the $\dfrac{dy}{dx}$ terms:

$$\frac{dy}{dx}(x + 2y) = -(2x + y) \implies \frac{dy}{dx} = -\frac{2x + y}{x + 2y}.$$

The procedure is always the same: differentiate, collect the $\frac{dy}{dx}$ terms on one side, factorise, divide.

This means every tangent and normal method from Ch. 14 now works on curves that are not functions.

EXAMPLE 15.9

Find the equation of the tangent to the curve $x^2 + xy + y^2 = 7$ at the point $(1, 2)$.
First confirm the point lies on the curve: $1 + 2 + 4 = 7$. From the previous example,

$$\frac{dy}{dx} = -\frac{2x + y}{x + 2y} = -\frac{2 + 2}{1 + 4} = -\frac{4}{5} \quad \text{at } (1, 2).$$

The tangent is $y - 2 = -\frac{4}{5}(x - 1)$, or $4x + 5y = 14$. Implicit differentiation supplies the gradient. The rest is the usual equation of a line through a given point.

YOUR TURN 15.5 Implicit Differentiation HL

11. Find $\dfrac{dy}{dx}$ for each curve.

(a) $x^2 + y^2 = 16$ [2]
(b) $xy = 12$ [2]
(c) $x^2 - y^2 = 9$ [2]
(d) $x^3 + y^3 = 6$ [2]

12. Each of these needs the product rule.

(a) Find $\dfrac{dy}{dx}$ for $x^2 + xy = 5$. [3]

(b) Find $\dfrac{dy}{dx}$ for $x^2y + y^2 = 4$. [3]

13.

(a) Find the gradient of $x^2 + y^2 = 25$ at the point $(4, -3)$. [3]

(b) Find the equation of the tangent to $xy = 4$ at the point $(2, 2)$. [4]

(c) On the circle $x^2 + y^2 = 25$, find the two points at which the tangent is horizontal. [3]

(d) Explain why implicit differentiation gives a gradient in terms of both x and y, while differentiating $y = f(x)$ gives one in terms of x alone. [2]

15.6 Related Rates HL

When two quantities are linked by an equation and both change over time, their rates of change are linked too. **Related rates** problems use the chain rule to connect those rates. You differentiate the equation that relates the quantities with respect to time.

The method has three steps. Write an equation relating the quantities. Differentiate every term with respect to t, so that each variable contributes its own rate. Then substitute the values known at the instant in question.

TIP *Substitute the given values only after differentiating. Putting $r = 5$ into the volume formula first turns a variable into a constant, and a constant has rate of change zero.*

EXAMPLE 15.10

Air is pumped into a spherical balloon at $100\,\text{cm}^3\text{s}^{-1}$. Find the rate at which the radius is increasing when $r = 5$ cm.

Volume and radius are related by $V = \frac{4}{3}\pi r^3$. Differentiate with respect to t:

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

Substitute $\dfrac{dV}{dt} = 100$ and $r = 5$:

$$100 = 4\pi(25)\frac{dr}{dt} \implies \frac{dr}{dt} = \frac{1}{\pi} \approx 0.318 \text{ cm s}^{-1}.$$

The rate of change of volume is constant, but the radius increases more slowly as the balloon grows, because the same volume is spread over a larger surface.

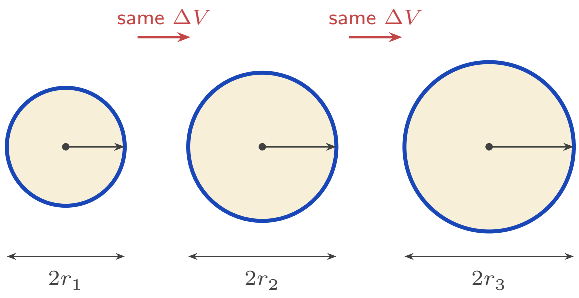

Figure 15.6 Equal volumes of air are added at each step, but the radius grows by less each time. The volume added is spread over the surface, and the surface area $4\pi r^2$ is itself growing, which is why $\frac{dr}{dt} = \frac{1}{4\pi r^2}\frac{dV}{dt}$ falls as r increases.

EXAMPLE 15.11

A 5 m ladder leans against a wall. The foot slides away from the wall at $0.5\,\mathrm{m\,s^{-1}}$. How fast is the top sliding down when the foot is 3 m from the wall?

Let x be the distance of the foot from the wall and y the height of the top. Then $x^2 + y^2 = 25$. Differentiate with respect to t:

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \implies \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}.$$

When $x = 3$, $y = \sqrt{25-9} = 4$. With $\dfrac{dx}{dt} = 0.5$:

$$\frac{dy}{dt} = -\frac{3}{4}(0.5) = -0.375 \text{ m s}^{-1}.$$

The negative sign means the top moves downward, as expected.

EXAMPLE 15.12

A water tank is an inverted cone with its vertex at the bottom, twice as tall as it is wide: at every level, the water surface has radius $r = \frac{h}{2}$, where h is the water depth. Water flows in at $2\,\mathrm{m^3\,min^{-1}}$. How fast is the depth rising when $h = 3$ m?

Write the volume in terms of h alone, using the shape constraint $r = \frac{h}{2}$:

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h = \frac{\pi h^3}{12}.$$

Differentiate with respect to time:

$$\frac{dV}{dt} = \frac{\pi h^2}{4}\frac{dh}{dt} \implies 2 = \frac{9\pi}{4}\frac{dh}{dt} \implies \frac{dh}{dt} = \frac{8}{9\pi} \approx 0.283 \text{ m min}^{-1}.$$

Remove r *before* differentiating. Differentiating an expression that still contains two linked variables is where most errors happen.

The algebra also shows the physics: the same inflow raises the level more slowly as the cone widens.

YOUR TURN 15.6 Related Rates HL

14.

(a) A square's side grows at $2\,\text{cm}\,\text{s}^{-1}$. Find the rate of change of its area when the side is 5 cm. [3]

(b) A circle's radius grows at $3\,\text{cm}\,\text{s}^{-1}$. Find the rate of change of its area when $r = 4$ cm. [3]

(c) A cube's side grows at $1\,\text{cm}\,\text{s}^{-1}$. Find the rate of change of its volume when the side is 10 cm. [3]

(d) A circle's radius increases at $0.4\,\text{cm}\,\text{s}^{-1}$. Find the rate of increase of its circumference. [2]

15.

(a) A spherical balloon is inflated at $50\,\text{cm}^3\text{s}^{-1}$. Find the rate of change of the radius when $r = 10$ cm. [3]

(b) A ladder 5 m long leans against a vertical wall. Its foot slides away from the wall at $2\,\text{m}\,\text{s}^{-1}$. Find the rate at which the top slides down when the foot is 3 m from the wall. [5]

(c) Water is poured into a cone at $20\,\text{cm}^3\text{s}^{-1}$. The cone's height is always twice its surface radius, so $V = \frac{2}{3}\pi r^3$. Find the rate at which r increases when $r = 5$ cm. [4]

Chapter 15 · Derivative Applications

Reflections

TOK The ball is motionless at the top of its flight, but only for an instant, and an instant has no duration. Is being motionless for no time at all a physical state, or an artefact of the mathematics? Does calculus describe motion, or replace it with something we can calculate?

TOK This chapter is full of conditions that are necessary but not sufficient: $f'(x) = 0$ holds at every smooth maximum, and $f''(x) = 0$ at every point of inflexion, yet neither condition guarantees one. What kind of knowledge is a necessary condition? Does knowing where something must be count as knowing where it is?

TOK The can that uses least metal has its height equal to its diameter, but real cans are taller and narrower, because the model ignores printing, shipping and the size of a hand. The mathematics is exact and the conclusion is still wrong. Does mathematically optimal always mean practically best?

IM The idea that nature minimises something appears across physics under many names. In seventeenth-century France, Pierre de Fermat proposed that light travels the path taking the least time, which explains refraction. Euler and Lagrange later generalised this into the calculus of variations and the *principle of least action*, which restates the whole of classical mechanics as an optimisation problem. The belief that the world is in some sense optimal has produced a great deal of physics.

RW The method behind modern machine learning is *gradient descent*, which is this chapter's idea at scale. A model has millions of adjustable numbers, and how wrong its predictions are is a function of those numbers; training means making that function as small as possible. With millions of variables the minimum cannot be found directly, so the computer does what a walker in fog would do: measure the slope where it stands, step down it, and repeat. As the slope flattens the steps shrink and the model settles near a minimum. That is $f' = 0$, applied billions of times.

EXT Implicit differentiation and related rates are the same idea twice. Both are the chain rule applied when the variables depend on something else: on x in the first case, on time in the second.

Carried further it becomes multivariable calculus, where a point has a different rate of change in every direction. Choosing the steepest of them is exactly what gradient descent does.

End-of-Chapter Problems — Chapter 15: Derivative Applications

1. For $f(x) = x^4$, show that $f''(0) = 0$ but that $(0,0)$ is *not* a point of inflexion. [4]

2. A stone dropped in a pond sends out a circular ripple whose radius grows at $0.5\,\mathrm{m\,s^{-1}}$. Find the rate at which the enclosed area is increasing when the radius is 8 m. [4]

3. A particle's velocity is $v(t) = 3t^2 - 12t + 9\,\mathrm{m\,s^{-1}}$, $t \geq 0$.

(a) Find the times when the particle is at rest. [2]
(b) Find the minimum velocity and the time at which it occurs. [3]

4. A company's profit from selling x units is $P(x) = -2x^2 + 120x - 1000$ dollars. Find the value of x that maximises the profit, justifying that it is a maximum, and state the maximum profit. [5]

5. The curve $\sin y + x^2 = 1$ passes through the point $(1,0)$. Find the gradient of the curve at that point. [5]

6. A curve has equation $x^2 + xy + y^2 = 3$.

(a) Find $\dfrac{dy}{dx}$ in terms of x and y. [3]
(b) Find the equation of the tangent at the point $(1,1)$. [3]

7. Water drains from an inverted cone whose radius always equals its height. The volume is $V = \frac{1}{3}\pi r^2 h$.

(a) Show that $V = \frac{1}{3}\pi h^3$. [2]
(b) Given the volume decreases at $2\,\mathrm{cm^3 s^{-1}}$, find the rate at which the depth h is falling when $h = 4$ cm. [4]

8. Consider $f(x) = x^3 - 3x$ on the closed interval $0 \leq x \leq 3$.

(a) Find the stationary point(s) in the interval and classify them. [3]
(b) Find the global maximum and minimum of f on the interval. [3]

9. Consider $g(x) = x + \dfrac{4}{x}$ on the interval $1 \leq x \leq 3$.

(a) Find the stationary point of g in the interval and classify it. [4]
(b) Find the maximum value of g on the interval, and state where it occurs. [2]

10. The curve C has equation $x^2 + y^2 - 4x + 2y = 15$.

(a) Show that the point $(0, 3)$ lies on C. [1]
(b) Use implicit differentiation to find $\dfrac{dy}{dx}$. [3]
(c) Find the equation of the tangent to C at $(0, 3)$. [2]

11. A street lamp is 5 m tall. A person of height 1.8 m walks away from it at $1.2\,\mathrm{m\,s^{-1}}$. Find the rate at which the length of the person's shadow is increasing. [6]

12. A water tank is an inverted cone in which the water surface radius is always $\frac{3}{4}$ of the depth h. Water is pumped in at $1.5\,\mathrm{m^3\,min^{-1}}$. Find the rate at which the depth is rising when $h = 2$ m. [6]

13. Find the point on the curve $y = \sqrt{x}$ closest to the point $(4, 0)$, and state the minimum distance. [6]

14. For the curve $x^2 + y^2 = 25$, use implicit differentiation twice to show that $\dfrac{d^2y}{dx^2} = -\dfrac{25}{y^3}$. [6]

15. A closed cylindrical can must hold $500\,\mathrm{cm^3}$.

(a) Show that the surface area is $S = 2\pi r^2 + \dfrac{1000}{r}$. [3]
(b) Find the radius that minimises S, and justify it is a minimum. [4]

16. A rectangular pen of area $200\,\mathrm{m^2}$ is built against a wall, needing fencing on three sides. Let the side perpendicular to the wall be x.

(a) Show that the length of fencing is $L = 2x + \dfrac{200}{x}$. [3]
(b) Find the dimensions that minimise the fencing used. [4]

17. The curve $y = x^3 + ax^2 + bx$ has stationary points at $x = 1$ and $x = -2$.

(a) Find the values of a and b. [4]
(b) Classify each stationary point. [3]

18. An open rectangular tank with a square base of side x must hold $32\,\mathrm{m^3}$.

(a) Show that the surface area (base + four sides) is $A = x^2 + \dfrac{128}{x}$. [3]
(b) Find the value of x that minimises the material used. [4]

19. An open-topped tank with a square base of side x cm is made from exactly $1200\,\mathrm{cm^2}$ of sheet metal.

(a) Show that the volume is $V = \dfrac{1200x - x^3}{4}$. [3]
(b) Find the dimensions that maximise the volume, and the maximum volume. [4]

20. A cone is inscribed in a sphere of radius R. Show that the cone of maximum volume has height $h = \frac{4R}{3}$. [7]

21. A 5 m ladder slides down a wall, its foot moving away at a constant $0.5\,\mathrm{m\,s^{-1}}$.

(a) Find the rate at which the top descends when the foot is 3 m from the wall. [3]

(b) Find the rate at which the area of the triangle formed by the ladder, wall, and ground is changing at that instant. [4]

22. A particle moves along the curve $y = x^2$ so that $\dfrac{dx}{dt} = 2$ at all times. Find the rate at which its distance from the origin is increasing as it passes through the point $(1, 1)$. [7]

23. Consider $f(x) = x^3 - 3x^2 - 9x + 5$.

(a) Find and classify the stationary points. [4]

(b) Find the point of inflexion. [2]

(c) State the intervals on which f is decreasing. [2]

24. A particle moves with displacement $s(t) = t^3 - 6t^2 + 9t$ metres, $t \geq 0$.

(a) Find the velocity and the times when the particle is at rest. [3]

(b) Find the acceleration when $t = 1$. [2]

(c) Determine whether the particle is speeding up or slowing down at $t = 1.5$. [3]

25. Consider $f(x) = x^3 - 6x^2 + 9x + 2$.

(a) Find and classify the stationary points. [5]

(b) Find the point of inflexion and show that the gradient there is not zero. [3]

26. A particle moves with displacement $s(t) = t^3 - 9t^2 + 24t - 10$ metres, for $0 \leq t \leq 6$ seconds.

(a) Find the times at which the particle is at rest. [3]

(b) Find its acceleration at each of those times. [2]

(c) Find the intervals of time during which the particle's *speed* is increasing. [3]

27. Consider $f(x) = xe^{-x}$.

(a) Find $f'(x)$ and the coordinates of the stationary point. [3]

(b) Determine its nature using the second derivative. [3]

(c) Find the x-coordinate of the point of inflexion. [2]

28. Consider $f(x) = x^4 - 4x^3$.

(a) Find $f'(x)$ and factorise it fully. [2]

(b) Find the coordinates of the stationary points. [3]

(c) Find $f''(x)$, and use it to classify the stationary point at $x = 3$. [3]

(d) Explain why the second derivative test fails at $x = 0$, and use the sign of f' to classify that point instead. [4]

(e) Find the coordinates of any points of inflexion. [3]

29. An open-topped box is made from a 12×12 cm square of card by cutting a square of side x cm from each corner and folding up the sides.

(a) Show that the volume is $V = x(12 - 2x)^2$, and state the possible values of x. [3]

(b) Find $\dfrac{dV}{dx}$ and show that it is zero at $x = 2$ and $x = 6$. [3]

(c) Explain why $x = 6$ must be rejected. [2]

(d) Find the maximum volume, justifying that it is a maximum. [4]

30. A particle moves in a straight line with displacement $s = t^3 - 6t^2 + 9t$ metres, for $t \geq 0$ seconds.

(a) Find expressions for the velocity and the acceleration. [3]

(b) Find the times at which the particle is momentarily at rest. [2]

(c) Find the displacement at each of those times. [2]

(d) Find the time at which the acceleration is zero, and state what is happening to the velocity at that instant. [3]

(e) Determine whether the particle is speeding up or slowing down at $t = 4$, giving a reason. [3]

31. The curve C has equation $x^3 + y^3 = 9xy$.

(a) Verify that the point $(2, 4)$ lies on C. [2]

(b) Use implicit differentiation to show that $\dfrac{dy}{dx} = \dfrac{x^2 - 3y}{3x - y^2}$. [4]

(c) Find the gradient of C at $(2, 4)$. [2]

(d) Find the equation of the tangent to C at $(2, 4)$. [3]

Challenge Questions

32. The shape of the best can. A closed cylindrical can must hold a fixed volume V. This investigation finds the shape that uses least metal, and shows that the answer does not depend on V.

(a) Show that the surface area is $S = 2\pi r^2 + \dfrac{2V}{r}$. [3]

(b) Find $\dfrac{dS}{dr}$, and show that S is stationary when $r^3 = \dfrac{V}{2\pi}$. [4]

(c) Verify, using the second derivative, that this gives a minimum. [3]
(d) Show that at this radius the height equals the diameter, whatever the value of V. [4]
(e) Real drink cans are taller and narrower than this. Suggest two costs the model leaves out. [2]

33. Closest approach. Finding the nearest point of a curve to a fixed point is an optimisation problem in disguise.

(a) Let $(x, \sqrt{x})$ be a point on $y = \sqrt{x}$ and let D be its distance from $(4, 0)$. Show that $D^2 = x^2 - 7x + 16$. [3]
(b) Explain why minimising D^2 gives the same answer as minimising D, and why using D^2 is easier. [3]
(c) Find the point on the curve closest to $(4, 0)$, and the least distance. [4]
(d) Show that at this closest point the line joining it to $(4, 0)$ is perpendicular to the tangent to the curve. [5]
(e) State the general result that part (d) illustrates. [2]

34. Quantifying curvature. A straight line does not bend; a small circle bends sharply. This investigation builds a number that measures how sharply a curve bends at a point. It is defined by

$$\kappa = \frac{|y''|}{\left(1 + (y')^2\right)^{3/2}},$$

and κ is called the *curvature*.

(a) Show that every straight line has $\kappa = 0$ at every point, and explain why the definition must give this. [3]
(b) For $y = x^2$, find κ as a function of x. Show that the curvature is greatest at the vertex, and find its value there. [4]
(c) Show that $\kappa \to 0$ as $x \to \pm\infty$ for $y = x^2$, and explain what this says about the shape of a parabola far from its vertex. [3]
(d) Use implicit differentiation on $x^2 + y^2 = R^2$ to show that a circle of radius R has $\kappa = \dfrac{1}{R}$ at every point. Explain why a constant curvature is exactly what you should expect. [5]
(e) The *radius of curvature* is defined as $\rho = \dfrac{1}{\kappa}$. Find ρ for $y = x^2$ at the origin, and state the circle that best matches the parabola there. [3]